

MOBEAST Preliminary Results

Keanu Rochette

2025-01-29

Load Libraries

Load function

Load Data

Codes of the treatment

Data clean up

Joining the data

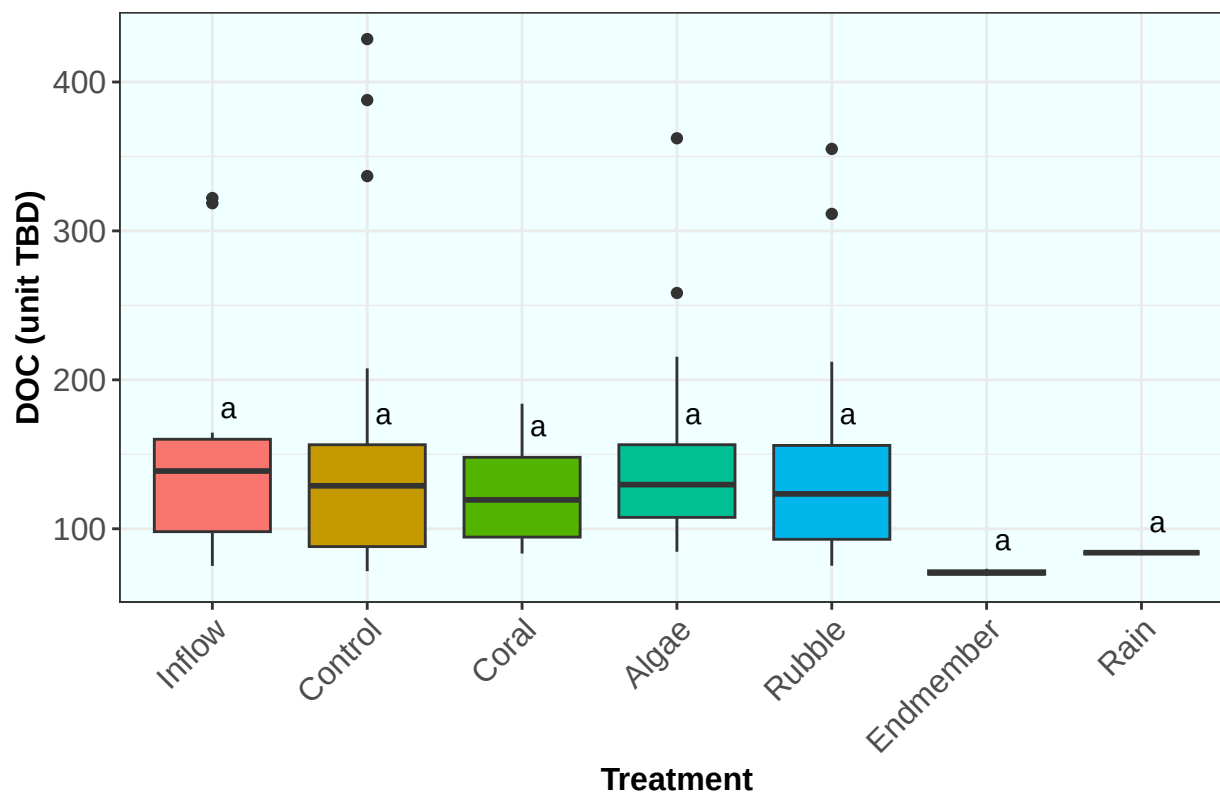
Data visualization

Run ANOVA on data

```
## [1] "indep var is a factor"
## [1] "dep var is numeric"
## [1] "Tk saved in environment"
```

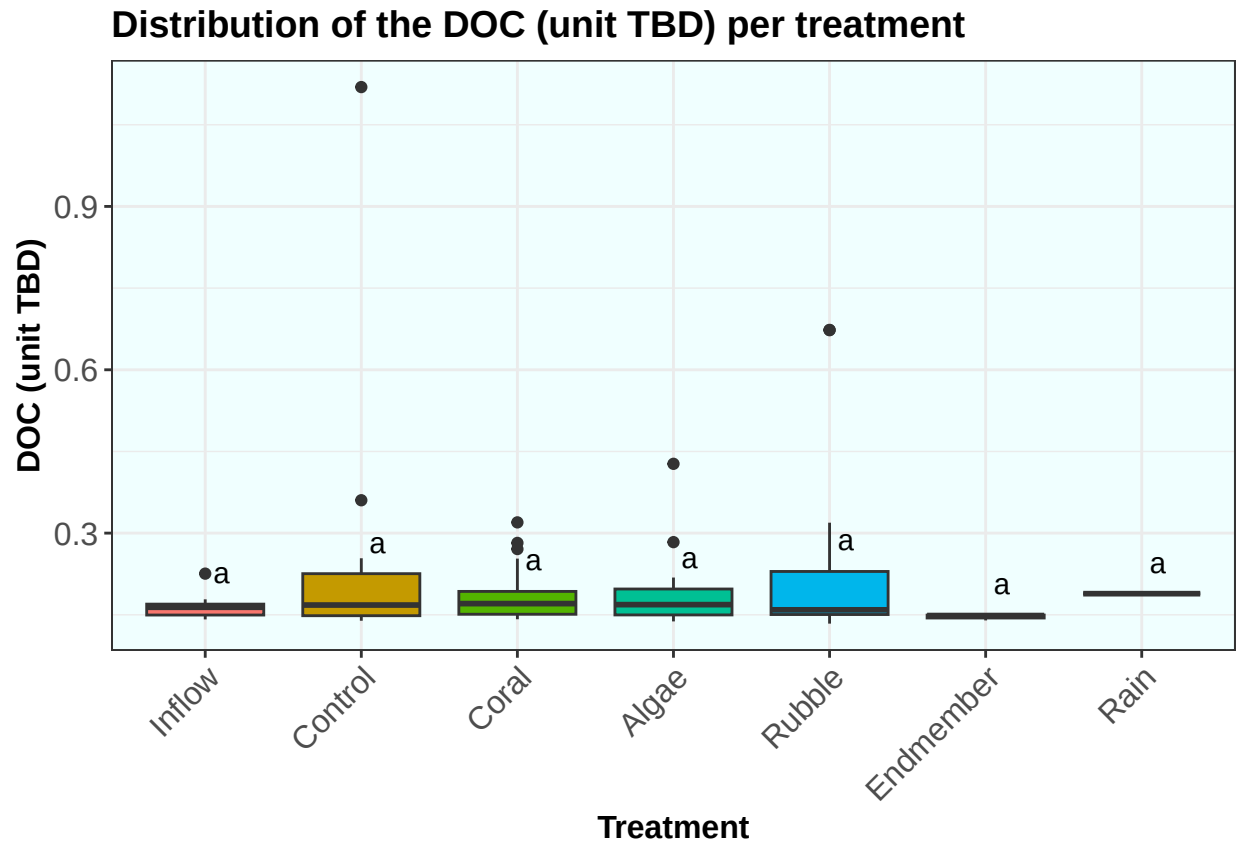
```
## # A tibble: 7 x 4
##   treatment    avg quant cld
##   <fct>      <dbl> <dbl> <chr>
## 1 Inflow     156.  160.  a
## 2 Control    151.  156.  a
## 3 Algae      146.  156.  a
## 4 Rubble     145.  156.  a
## 5 Coral      122.  148.  a
## 6 Rain        83.9  83.9  a
## 7 Endmember  70.7  71.8  a
```

Distribution of the DOC (unit TBD) per treatment



```
## [1] "indep var is a factor"
## [1] "dep var is numeric"
## [1] "Tk saved in environment"
```

```
## # A tibble: 7 x 4
##   treatment  avg quant cld
##   <fct>      <dbl> <dbl> <chr>
## 1 Rubble    0.225 0.230 a
## 2 Control   0.224 0.226 a
## 3 Rain      0.189 0.189 a
## 4 Algae     0.186 0.197 a
## 5 Coral     0.186 0.193 a
## 6 Inflow    0.165 0.170 a
## 7 Endmember 0.146 0.150 a
```



Linear Model

```
##
## Call:
## lm(formula = npoc_u_m ~ treatment, data = DOC)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -81.23  -43.08  -17.48   14.05  278.01
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      156.30      21.24   7.360 4.64e-11 ***
## treatmentControl    -5.49      25.64  -0.214  0.8309
## treatmentCoral     -33.80      25.64  -1.318  0.1904
## treatmentAlgae     -10.09      25.64  -0.394  0.6948
## treatmentRubble    -11.65      25.82  -0.451  0.6529
## treatmentEndmember -85.60      45.88  -1.866  0.0649 .
## treatmentRain      -72.37      73.56  -0.984  0.3276
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 70.43 on 103 degrees of freedom
## (5 observations deleted due to missingness)
## Multiple R-squared:  0.05903,    Adjusted R-squared:  0.004216
```

```
## F-statistic: 1.077 on 6 and 103 DF,  p-value: 0.3812
```

```
## Anova Table (Type II tests)
```

```
##
```

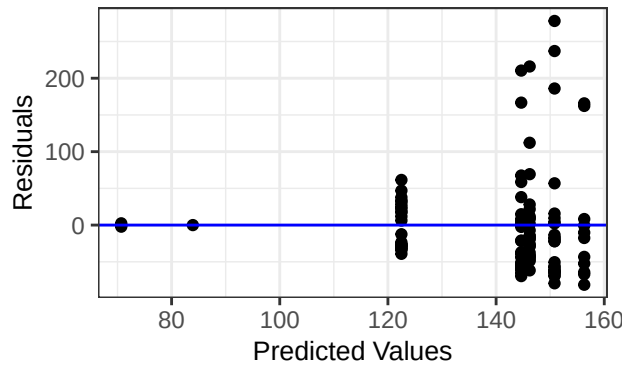
```
## Response: npoc_u_m
```

```
##           Sum Sq Df F value Pr(>F)
```

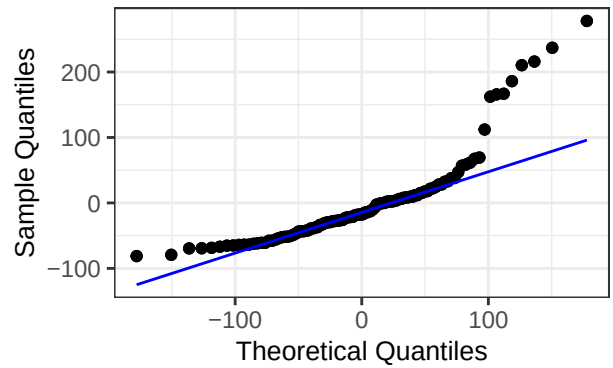
```
## treatment  32055  6  1.0769 0.3812
```

```
## Residuals 510967 103
```

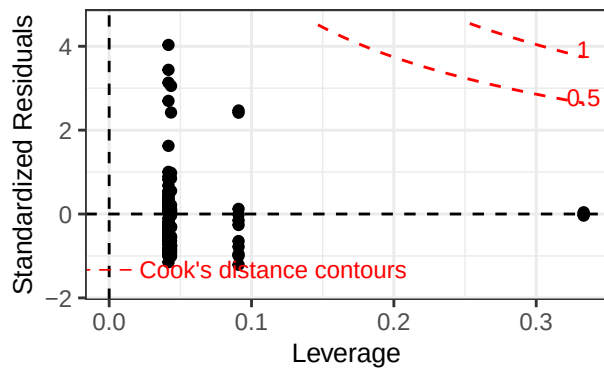
Residual Plot



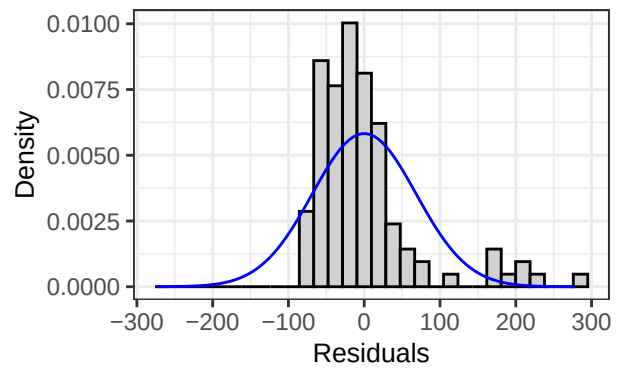
Q-Q Plot

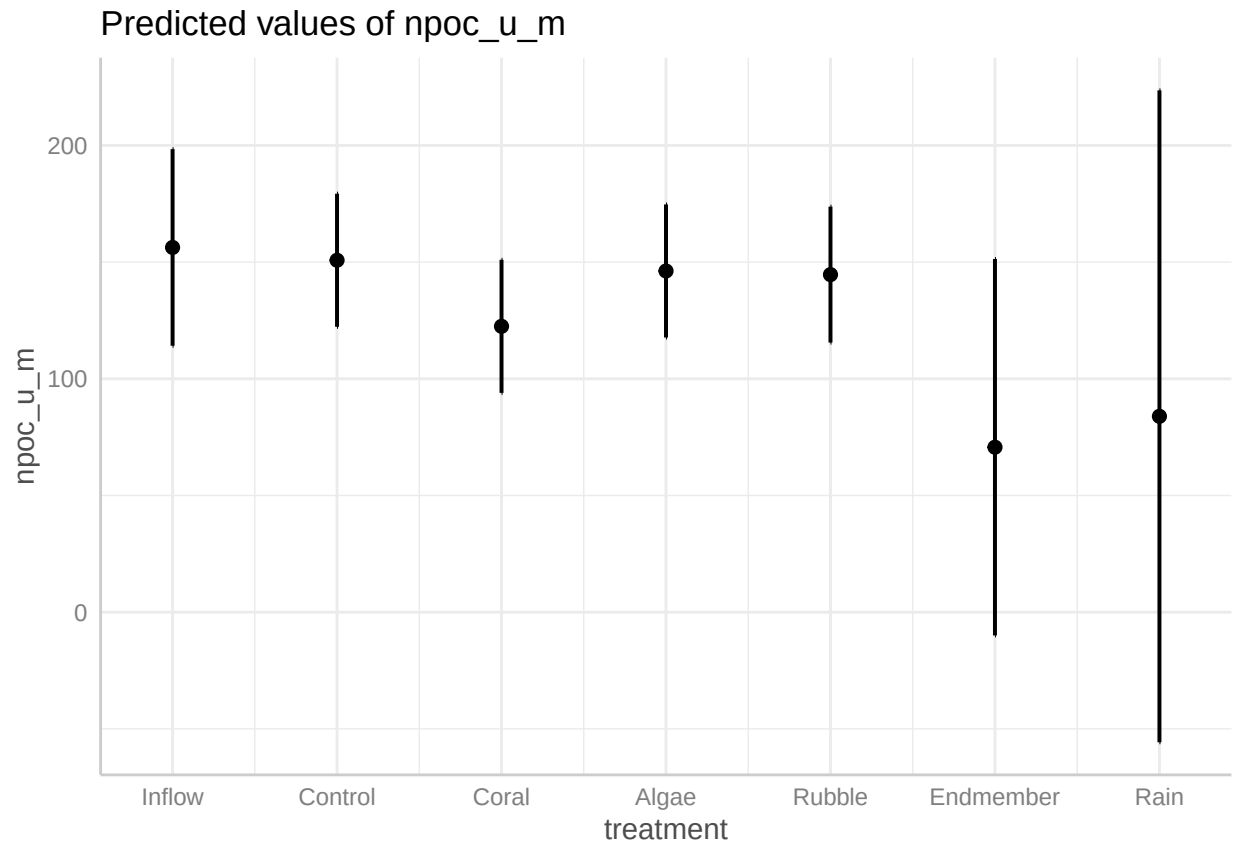


Residual-Leverage Plot



Histogram





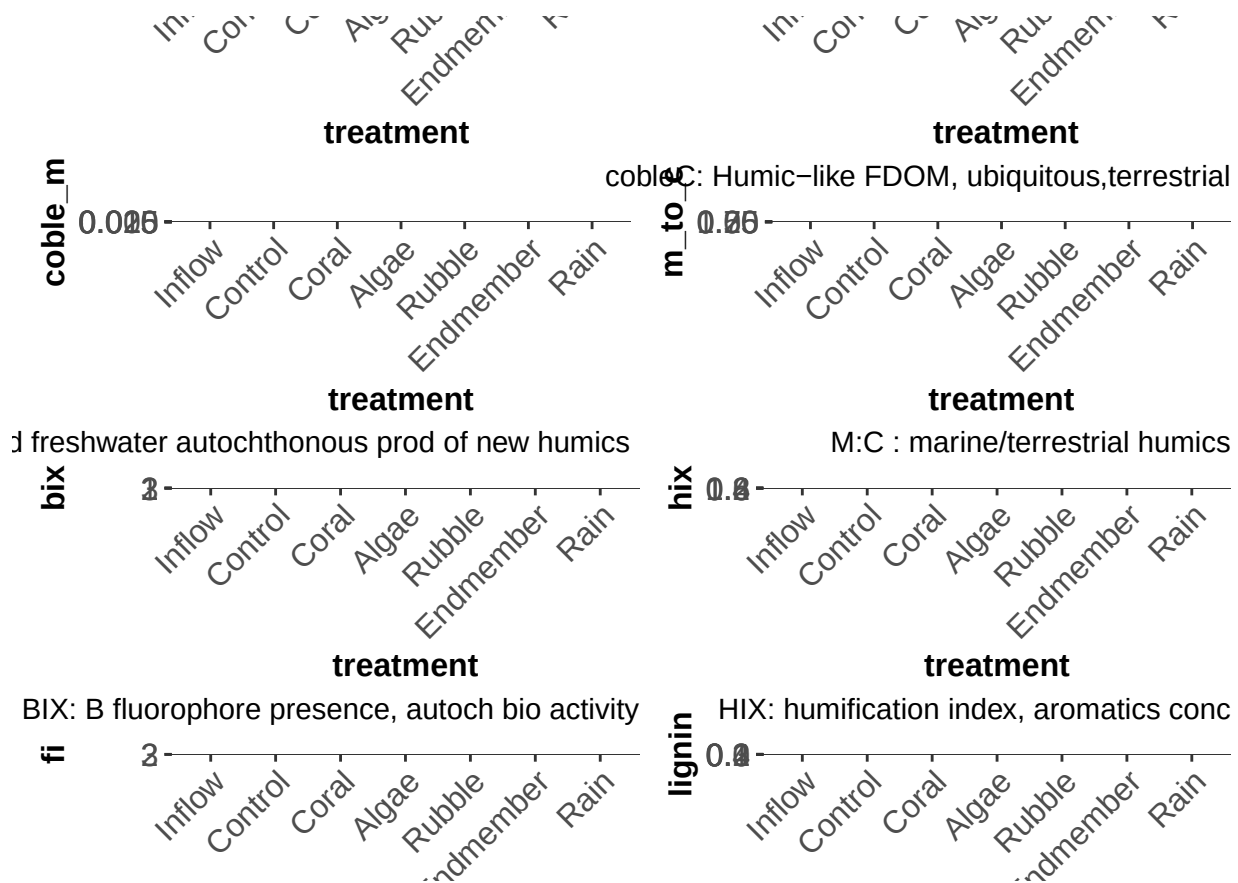
fDOM

```
## [1] "indep var is a factor"
## [1] "dep var is numeric"
## [1] "Tk saved in environment"

## # A tibble: 7 x 4
##   treatment      avg  quant cld
##   <fct>         <dbl> <dbl> <chr>
## 1 Rain          0.0218  0.0218 a
## 2 Algae          0.0103  0.0125 b
## 3 Rubble         0.00969 0.0117 b
## 4 Inflow         0.00916 0.0101 bc
## 5 Control        0.00914 0.0119 bc
## 6 Coral          0.00852 0.0111 bc
## 7 Endmember      0.00479 0.00578 c

## List of 1
## $ plot.caption:List of 11
## ..$ family      : NULL
## ..$ face         : NULL
## ..$ colour       : NULL
## ..$ size         : num 11
## ..$ hjust        : NULL
## ..$ vjust        : NULL
```

```
## ..$ angle      : NULL
## ..$ lineheight  : NULL
## ..$ margin      : NULL
## ..$ debug       : NULL
## ..$ inherit.blank: logi FALSE
## ..- attr(*, "class")= chr [1:2] "element_text" "element"
## - attr(*, "class")= chr [1:2] "theme" "gg"
## - attr(*, "complete")= logi FALSE
## - attr(*, "validate")= logi TRUE
```



Distribution of the fDOM index per treatment

